

FACTORING TRINOMIALS — Student Worksheet

$$x^2 + bx + c \quad (a = 1) \quad \cdot \quad \text{The X-Puzzle} \quad \cdot \quad \text{TEKS A.10E}$$

Name: _____ Date: _____ Class: _____

Learning Objective: I can factor a trinomial $x^2 + bx + c$ ($a = 1$) into two binomials.

Goal: Factor at least 3 of the 4 exit-ticket problems correctly.

DO NOW Warm-up — multiply these out. (Factoring is just this in reverse!)

Use FOIL. Show every term.

1) $(x + 3)(x + 5) =$

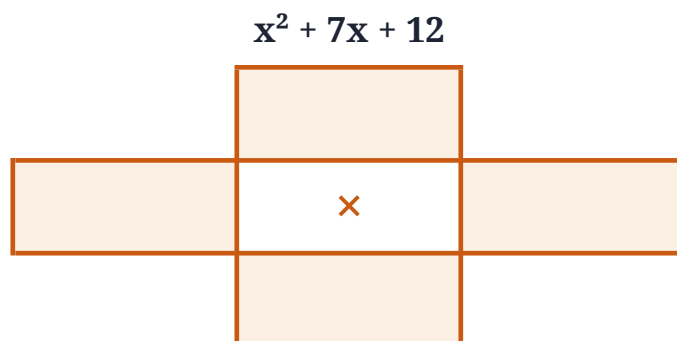
2) $(x + 6)(x - 2) =$

NOTES — THE X-PUZZLE I Do · fill this in as we model $x^2 + 7x + 12$

The 3 questions, every time:

1. GCF? (factor it out first)
2. What two numbers MULTIPLY to c (the top)?
3. Of those, which pair ADDS to b (the bottom)?

Stamp: c is the PRODUCT. b is the SUM.



Sign chart — your two numbers decide the product (c) and the sum (b):

Your two numbers	Product $\rightarrow c$	Sum $\rightarrow b$
$+, +$	$+$	$+$
$-, -$	$+$	$-$
$+, -$	$-$	$+$ e.g. $+9, -3 \rightarrow +6$
$-, +$	$-$	$-$ e.g. $-7, +5 \rightarrow -2$

Opposite signs $\rightarrow$ product (c) is $-$, and the sum (b) takes the sign of the bigger number.

STUCK? Don't freeze — climb the ladder:

1. List every pair that MULTIPLIES to c.
2. Circle the pair that ADDS to b.
3. Use the sign chart to fix the $+$ / $-$ signs.
4. Still stuck? Raise your hand — I'll come.

WE DO — together *You call the numbers. Convince your partner of the signs.***① Factor: $x^2 + 9x + 20$**

	$\times$	

top = product (c) · bottom = sum (b) · left & right = the two numbers

Factored form: $(x \text{ } \underline{\hspace{1cm}})(x \text{ } \underline{\hspace{1cm}})$ **Check (FOIL):**

② Factor: $x^2 + 2x - 15$

	$\times$	

top = product (c) · bottom = sum (b) · left & right = the two numbers

Factored form: $(x \text{ } \underline{\hspace{1cm}})(x \text{ } \underline{\hspace{1cm}})$ **Check (FOIL):**

YOU DO — on your own *Silent · 3 minutes · show your X-puzzle · whiteboards up at the end***③ Factor: $x^2 - 7x + 12$**

	$\times$	

top = product (c) · bottom = sum (b) · left & right = the two numbers

Factored form: $(x \text{ } \underline{\hspace{1cm}})(x \text{ } \underline{\hspace{1cm}})$ **Check (FOIL):**

④ **Factor:** $x^2 + 3x - 10$



top = product (c) · bottom = sum (b) · left & right = the two numbers

Factored form: $(x \quad)(x \quad)$ **Check (FOIL):** _____

CHALLENGE ZONE *Finished early? Climb as high as you can — prove every answer by expanding (FOIL).*

① $x^2 - 13x + 36 = (x \quad)(x \quad)$ ← bigger numbers

② $x^2 - 49 = (x \quad)(x \quad)$ ← where is the middle term?

③ $2x^2 + 7x + 3 = (\quad x \quad)(x \quad)$ ← careful — the x^2 has a 2!

④ **Create your own:** write a trinomial that factors to $(x + 6)(x - 4)$ — then expand to prove it:

EXIT TICKET — Demonstration of Learning

This is collected. Factor all four. Show your work.

Name: _____ Class: _____

Two quick things so I can get to know you (takes 10 seconds):

Favorite hobby: _____ Favorite game / sport: _____

1) Factor: $x^2 + 8x + 15$

	×	

Answer: $(x \text{ } ______)(x \text{ } ______)$

2) Factor: $x^2 - 9x + 20$

	×	

Answer: $(x \text{ } ______)(x \text{ } ______)$

3) Factor: $x^2 + x - 12$

	×	

Answer: $(x \text{ } ______)(x \text{ } ______)$

4) Factor: $x^2 - 4x - 21$

	x	

Answer: (x _____)(x _____)

Self-check: Does each pair MULTIPLY to c and ADD to b ? Multiply back to be sure.